

A MAJOR PROJECT REPORT ON

AI-DRIVEN INTELLIGENT GROCERY RETAIL

SYSTEM USING MACHINE LEARNING

Submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE & ENGINEERING (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

SUBMITTED BY:

Shashikant Shukla (Roll No. / Moodle ID: [STUDENT_1_MOODLE_ID], PRN:

[STUDENT_1_PRN])

Om Dubey (Roll No. / Moodle ID: [STUDENT_2_MOODLE_ID], PRN: [STUDENT_2_PRN])

Shreyash Wadalkar (Roll No. / Moodle ID: [STUDENT_3_MOODLE_ID], PRN:

[STUDENT_3_PRN])

[STUDENT_4_NAME — DO NOT GUESS] (Roll No. / Moodle ID:

[STUDENT_4_MOODLE_ID], PRN: [STUDENT_4_PRN])

UNDER THE GUIDANCE OF:

[PROJECT_GUIDE_NAME_AND_TITLE]

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (AIML)

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

(Approved by AICTE, Recognized by Govt. of Maharashtra, Affiliated to University of Mumbai)

Survey No. 12, Opp. Hypercity Mall, Kasarvadavali, Ghodbunder Road, Thane (West) –

400 615

ACADEMIC YEAR: 2025–2026

CERTIFICATE OF APPROVAL

This is to certify that the project entitled "AI-Driven Intelligent Grocery Retail System Using Machine Learning" is a bonafide work carried out by Shashikant Shukla ([STUDENT_1_MOODLE_ID]), Om Dubey ([STUDENT_2_MOODLE_ID]), Shreyash Wadalkar ([STUDENT_3_MOODLE_ID]), and [STUDENT_4_NAME — DO NOT GUESS] ([STUDENT_4_MOODLE_ID]) in partial fulfillment of the requirements for the award of the Degree of Bachelor of Engineering in Computer Science & Engineering (Artificial Intelligence & Machine Learning) from the University of Mumbai for the academic year 2025–2026.

This project has been completed under our supervision and guidance, and to the best of our knowledge, the work presented herein has not been submitted elsewhere for the award of any other degree or diploma.

[PROJECT_GUIDE_NAME]

Project Guide

Dept. of CSE (AIML), APSIT

[PROJECT_COORDINATOR_NAME]

Project Coordinator

Dept. of CSE (AIML), APSIT

[HOD_NAME]

Head of Department

Dept. of CSE (AIML), APSIT

Dr. U. V. Bhosale

Principal

A. P. Shah Institute of Technology

PROJECT REPORT APPROVAL

This project report entitled "AI-Driven Intelligent Grocery Retail System Using Machine Learning" submitted by Shashikant Shukla ([STUDENT_1_MOODLE_ID]), Om Dubey ([STUDENT_2_MOODLE_ID]), Shreyash Wadalkar ([STUDENT_3_MOODLE_ID]), and [STUDENT_4_NAME — DO NOT GUESS] ([STUDENT_4_MOODLE_ID]) is approved for the degree of Bachelor of Engineering in Computer Science & Engineering (Artificial Intelligence & Machine Learning) by the Board of Examiners.

Board of Examiners:

1. Internal Examiner: _____ Signature: _____ Date: _____

2. External Examiner: _____ Signature: _____ Date: _____

STUDENT DECLARATION

We hereby declare that the project report entitled "AI-Driven Intelligent Grocery Retail System Using Machine Learning" submitted by us to the Department of Computer Science & Engineering (Artificial Intelligence & Machine Learning), A. P. Shah Institute of Technology, Thane, affiliated with the University of Mumbai, is a record of original engineering work done by us under the supervision of [PROJECT_GUIDE_NAME_AND_TITLE].

We further declare that:

1. The empirical results, algorithm implementations, database schemas, and architectural designs documented herein are authentic and derived strictly from our implemented full-stack codebase and empirical test harness.
2. The references cited in this report represent real, peer-reviewed scientific publications verified through the IEEE Xplore Digital Library.
3. This work has not been previously submitted to any other university or institute for the award of any degree, diploma, or certificate.

Candidates' Signatures:

1. Shashikant Shukla: _____ Date: _____

2. Om Dubey: _____ Date: _____

3. Shreyash Wadalkar: _____ Date: _____

4. [STUDENT_4_NAME — DO NOT GUESS]: _____ Date: _____

ABSTRACT

The rapid growth of on-demand quick-commerce and urban dark-store fulfillment has created severe operational challenges for grocery retail systems. Modern grocery retailing is characterized by thin operating margins (2–5%), high perishable product spoilage (15–25%), volatile intraday customer demand, and tight delivery windows (10–30 minutes). Conventional retail platforms typically decouple customer-facing personalization from backend supply chain and order-fulfillment operations. This separation results in frequent stockouts, excessive holding costs, inefficient warehouse picker routing, and uncoordinated vehicle fleet dispatching. Furthermore, emerging machine learning implementations frequently suffer from methodological data leakage, unconstrained dynamic pricing instability, and high inference latency that exceeds real-time web application service level agreements.

To address these challenges, this project presents FreshCart AI, an integrated, resilient, and leak-free intelligent grocery retail platform that unifies predictive machine learning with combinatorial operations research within a high-performance, two-tier microservice architecture. The platform combines four predictive machine learning modules with three mathematical operations research optimizers: (1) a weighted Hybrid Collaborative Filtering and TF-IDF Content-Based Recommendation Engine ($\alpha=0.60$); (2) a multi-step recursive SARIMAX(1,1,1)x(1,0,1)⁷ SKU demand forecasting model with calendar and promotional exogenous regressors; (3) an econometric Log-Log Ordinary Least Squares (OLS) dynamic pricing engine constrained within safety bounds ($\pm 25\%$); (4) a cost-sensitive Random Forest transaction fraud risk classifier; (5) an automated Continuous Review (r, Q) multi-item inventory optimizer implementing Wilson Economic Order Quantity (EOQ) and Gaussian stochastic safety stock; (6) a dark-store 2D Traveling Salesperson Problem (TSP) picker walk path optimizer combining greedy Nearest-Neighbor initialization with intra-tour 2-Opt local search; and (7) a Capacitated Vehicle Routing Problem (CVRP) last-mile delivery fleet dispatch solver utilizing Clarke-Wright Savings clustering and intra-route 2-Opt smoothing.

The system is deployed on a two-tier architecture linking a Node.js Express application server with an asynchronous Python FastAPI machine learning microservice. To support operational resilience, an AI Gateway client implements an asynchronous non-blocking dispatcher with a strict 1500ms circuit breaker timeout and graceful fallback to in-process

heuristic algorithms. Rigorous empirical holdout evaluation demonstrates consistent system performance across the experimental test harness: the hybrid recommendation engine achieves an F1@10 of 0.5027 and an NDCG@10 of 0.9790; recursive SARIMAX demand forecasting yields an out-of-sample Root Mean Squared Error (RMSE) of 5.83 units (MAPE = 2.50%); dynamic pricing simulation produces a model-based daily revenue lift of +22.21% ($p < 0.001$) under Constant Elasticity of Demand (CED) assumptions; transaction fraud scoring achieves an ROC-AUC of 0.6087 on an uncorrupted holdout dataset with zero synthetic target leakage; in a 365-day simulation benchmark, continuous review (r, Q) inventory optimization reduces simulated holding and ordering costs by 87.64% while elevating the cycle service level to 99.88%; dark-store picker walking distance is reduced by 37.48% (achieving a 0.09% average gap versus exact brute-force solutions); and last-mile delivery fleet travel distance is reduced by 61.62% in benchmark scenarios with vehicle capacity utilization increasing from 38.4% to 82.9%. In local development benchmarks, all microservice endpoints execute with sub-25ms p95 latency. The full system is validated by an automated regression test suite comprising 113 passed assertions and 56 master audit checks, confirming the stability, scalability, and academic defensibility of the integrated platform.

Keywords: Intelligent Grocery Retail, Hybrid Recommendation System, SARIMAX Demand Forecasting, Dynamic Pricing Elasticity, Random Forest Fraud Detection, Continuous Review Inventory Policy, 2D TSP Warehouse Order Picking, Capacitated Vehicle Routing Problem (CVRP), Microservice Architecture, Fault-Tolerant Circuit Breaker.

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We express our profound gratitude to our project guide, [PROJECT_GUIDE_NAME_AND_TITLE], Department of Computer Science & Engineering (AIML), A. P. Shah Institute of Technology, Thane, for their constant encouragement, insightful guidance, and constructive critique throughout the conceptualization, development, and evaluation of this major engineering project.

We extend our sincere thanks to [PROJECT_COORDINATOR_NAME], Project Coordinator, and [HOD_NAME], Head of the Department of Computer Science & Engineering (AIML), for providing excellent departmental facilities, high-performance computing resources, and administrative support.

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Finally, we express our heartfelt appreciation to our parents, family members, and peers for their unconditional support, patience, and motivation throughout the course of our undergraduate engineering curriculum.

TABLE OF CONTENTS

Certificate of Approval	ii
Project Report Approval	iii
Student Declaration	iv
Abstract	v
Acknowledgements	vi
List of Figures	viii
List of Tables	ix
List of Abbreviations	x
CHAPTER 1 — INTRODUCTION	1
CHAPTER 2 — LITERATURE SURVEY	8
CHAPTER 3 — EXISTING SYSTEM AND LIMITATIONS	14
CHAPTER 4 — PROBLEM STATEMENT, OBJECTIVES AND SCOPE	17
CHAPTER 5 — PROPOSED SYSTEM & TECHNICAL ARCHITECTURE	20
CHAPTER 6 — EXPERIMENTAL SETUP & METHODOLOGY	50
CHAPTER 7 — RESULTS AND DISCUSSION	56
CHAPTER 8 — CONCLUSION AND FUTURE WORK	74
REFERENCES	77
APPENDICES	79

LIST OF FIGURES

Figure 5.1: Customer Storefront PWA Home View (31 Seeded Grocery SKUs)	22
Figure 5.2: Bilingual NLP Smart Search Interface (Query: 'organic milk')	23
Figure 5.3: Personalized Top-K Hybrid Recommendation Carousel	24
Figure 5.4: Active Shopping Cart Drawer & Checkout View	25
Figure 5.5: Executive Admin Management Dashboard Overview	26
Figure 5.6: Interactive 30-Day SARIMAX Demand Forecasting Visualizer	27
Figure 5.7: Econometric Dynamic Pricing & Price Elasticity Simulator	28
Figure 5.8: Live Customer Orders Feed & Real-Time Fraud Risk Scoring	29
Figure 5.9: Automated Inventory & Stock Risk Alert Table	30
Figure 5.10: Last-Mile Delivery Route Optimizer (VRP / 2-Opt Map)	31
Figure 5.11: Python AI Microservice Gateway Health Status Badge	32
Figure 7.1: Out-of-Sample Demand Forecasting — Actual vs. Predicted (30-Day Horizon)	58
Figure 7.2: Empirical Price Elasticity of Demand Curves across Categories	61
Figure 7.3: Transaction Fraud Risk Scoring — Receiver Operating Characteristic (ROC)	64
Figure 7.4: Annual Inventory Holding, Ordering & Total Cost Comparison	67
Figure 7.5: Dark Store Warehouse Picker Walk Distance (100 Order Batches)	70
Figure 7.6: Last-Mile Delivery Fleet Distance & Vehicle Capacity Utilization (100 Instances)	72
Figure 7.7: Local Gateway & Solver Latency Benchmarks (Mean vs. p95 ms)	73

LIST OF TABLES

Table 2.1: Comparative Literature Survey Matrix of Recent IEEE Research (2023–2026)	11
Table 6.1: Hardware and Execution Environment Specifications	51
Table 6.2: Software Dependency Stack and Versions	52
Table 6.3: Dataset Characteristics and Statistical Distributions	54
Table 7.1: Personalized Top-K Recommendation Holdout Evaluation	57
Table 7.2: 30-Day SKU Demand Forecasting Accuracy	59
Table 7.3: Dynamic Pricing & Price Elasticity Simulation Results	62
Table 7.4: Transaction Fraud Risk Scoring Holdout Classification	65
Table 7.5: Continuous Review (r, Q) Inventory Simulation Benchmarks	68
Table 7.6: Dark Store 2D TSP Picker Walk Optimization Benchmarks	70
Table 7.7: Capacitated Vehicle Routing Problem (CVRP) Delivery Logistics Benchmarks	72
Table 7.8: Empirical Gateway and Solver Latency Benchmarks (p95)	73
Table 7.9: Automated Verification & Code Quality Regression Test Suite Summary	74

LIST OF ABBREVIATIONS

Abbreviation	Full Term / Description
AI	Artificial Intelligence
AIML	Artificial Intelligence and Machine Learning
API	Application Programming Interface
AUC	Area Under the Curve
CED	Constant Elasticity of Demand
CF	Collaborative Filtering
CVRP	Capacitated Vehicle Routing Problem
EOQ	Economic Order Quantity
F1	F1 Score (Harmonic Mean of Precision and Recall)
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
NDCG	Normalized Discounted Cumulative Gain
OLS	Ordinary Least Squares
OR	Operations Research
PWA	Progressive Web Application
RBAC	Role-Based Access Control
RMSE	Root Mean Squared Error
ROC	Receiver Operating Characteristic
ROP	Reorder Point
SARIMAX	Seasonal Autoregressive Integrated Moving Average with Exogenous Regressors
SKU	Stock Keeping Unit
TF-IDF	Term Frequency-Inverse Document Frequency
TSP	Traveling Salesperson Problem
VRP	Vehicle Routing Problem

CHAPTER 1 — INTRODUCTION

1.1 Background and Industry Context

The global retail sector is undergoing a profound paradigm shift driven by digital commerce, mobile applications, and rapid-fulfillment logistics. In urban consumer markets, customer expectations have evolved from multi-day delivery windows to hyper-local "quick commerce," where grocery orders are expected to be picked, packed, and delivered within 10 to 30 minutes. Grocery retailing, however, is structurally constrained by razor-thin profit margins (typically 2–5%), extreme demand volatility across perishable product lines, short inventory shelf lives, and high fulfillment labor costs.

To maintain operational viability in this hyper-competitive environment, retail platforms must transition from static, manual operational models to intelligent, automated systems. Modern artificial intelligence (AI) and machine learning (ML) paradigms offer unprecedented opportunities to analyze customer interaction clickstreams, predict granular SKU-level sales trends, adapt pricing to market demand elasticity, and identify fraudulent transactions. Concurrently, operations research (OR) methodologies provide the mathematical foundation required to optimize continuous inventory replenishment, minimize dark-store picker walking paths, and dispatch multi-vehicle delivery fleets along optimal geographical routes.

1.2 Motivation

Traditional grocery systems and conventional e-commerce platforms suffer from severe operational inefficiencies:

1. **Perishable Inventory Losses:** Perishable items (fruits, dairy, fresh vegetables) incur severe spoilage losses (15–25%) when store procurement relies on static rule-of-thumb ordering rather than predictive demand forecasting.
2. **Stockouts and Lost Revenue:** Inaccurate sales estimates lead to stockouts on fast-moving consumer goods, directly degrading customer satisfaction and lifetime retention.
3. **Dark-Store Picking Bottlenecks:** Order pickers inside micro-fulfillment centers ("dark stores") manually traverse aisles in sequential order, resulting in physical fatigue, long order-assembly times, and missed 10-minute dispatch windows.
4. **Logistical Inefficiencies:** Last-mile couriers operate uncoordinated, single-order radial runs, leading to low vehicle payload utilization, high fuel expenses, and excessive fleet

mileage.

5. Static Pricing Inefficiencies: Retailers lack real-time mechanisms to adjust prices dynamically based on estimated category price elasticity (E_d) and perishable shelf-life decay.

1.7 Functional and Technical Objectives

The project encompasses eight primary engineering objectives:

1. Personalization: Implement a high-precision Top-K Hybrid Recommendation Engine combining Collaborative Filtering with TF-IDF Content matching.
2. Demand Forecasting: Implement a leak-free 30-day recursive daily SKU demand forecaster utilizing SARIMAX(1,1,1)x(1,0,1)⁷ with calendar and promotional regressors.
3. Dynamic Pricing: Formulate an econometric Log-Log OLS price elasticity optimizer with bounded $[\pm 25\%]$ safety guardrails.
4. Fraud Detection: Build a cost-sensitive Random Forest transaction risk scoring classifier operating under severe class imbalance.
5. Inventory Optimization: Develop an automated Continuous Review (r, Q) inventory optimizer computing Wilson EOQ and stochastic safety stock.
6. Warehouse Picking: Engineer a dark-store 2D TSP picker walk path optimizer combining Nearest-Neighbor initialization with 2-Opt local search.
7. Delivery Logistics: Implement a Capacitated Vehicle Routing Problem (CVRP) solver combining Clarke-Wright Savings clustering with 2-Opt smoothing.
8. Resilient Integration: Design a two-tier Node.js <-> Python FastAPI architecture with a 1500ms circuit breaker and resilient in-process fallback hierarchy.

CHAPTER 2 — LITERATURE SURVEY

2.1 Overview of Surveyed Literature

A systematic literature survey was conducted exclusively across peer-reviewed publications indexed in the IEEE Xplore Digital Library published between 2023 and 2026. A total of 15 recent IEEE publications were selected and analyzed across e-commerce recommendations ([1]–[3]), demand forecasting ([4]–[6]), dynamic pricing ([7]–[8], [11]), transaction fraud detection ([9]–[10]), operations research logistics ([13]–[15]), and edge retail architectures ([11]–[12]).

Table 2.1: Comparative Literature Survey Matrix of Recent IEEE Research (2023–2026)

Year	Paper Title & Citation	Domain	Algorithm / Method	Dataset	Key Finding	Identified Gap
2024	Enhancing Recommender Systems: Hybrid Approach [1]	Recommendation	Hybrid Sentiment + CF	E-Commerce Logs	CF + item sentiment boosts precision & recall.	High text parsing compute latency (>50ms).
2024	Landscape of Hybrid Recommendation Systems [2]	Hybrid E-Comm	Systematic Review	120+ Retail Systems	Weighted linear hybrids yield top ranking stability.	Lacks link to real-time inventory stock.
2023	Deep Learning-Based Recommendation System [3]	Deep Recs	Autoencoders, GNNs	Retail Benchmarks	Non-linear interaction modeling.	Heavy matrix operations exceed edge CPU SLAs.
2024	Demand Forecasting in SCM for Rossmann [4]	Demand Forecast	Weather/Calendar DL	Rossmann Stores	Exogenous indicators cut sales RMSE.	Risk of lookahead lag leakage in naive setups.
2023	Demand Forecasting for Multi-channel Retail [5]	Retail Forecast	CatBoost, XGBoost	Daily Sales Stream	30-day recursive daily sales horizon.	Evaluated in isolation without ROP triggers.
2024	Smart Retail: Demand & Inventory Optimization [6]	Smart Retail	Time-Series ML + EOQ	Inventory Streams	Sales forecasts cut stockout rate by >40%.	Static safety buffer without variance

						model.
2024	Dynamic Pricing: Trends & New Frontiers [7]	Dynamic Pricing	Pricing Review	Digital Retail	Unconstrained pricing causes consumer churn.	Qualitative review lacking closed-form solvers.
2024	AI & ML for Dynamic Pricing Strategies [8]	Revenue Strategy	Demand Elasticity (Ed)	Sales Transaction Logs	Elasticity curve fitting maximizes revenue.	Slow convergence in high-concurrency checkouts.
2024	Credit Card Fraud Detection Ensemble [9]	Fraud Detection	Random Forest Ensemble	POS Transactions	Random Forest outperforms single trees & linear.	Requires feature scaling to prevent velocity leak.
2024	Deep Learning for Credit Card Fraud [10]	Imbalanced Fraud	Deep Learning Review	European CC Logs	ROC-AUC is the true metric on <1% fraud.	Neural classifiers introduce >50ms latency.
2024	Smart Retail: Demand, Price & Inventory [11]	Unified Retail	Integrated Pipeline	Store Enterprise Logs	Unified demand + pricing + inventory loop.	Lacks warehouse picking & vehicle routing.
2025	Smart Retail Solutions: Edge Computing [12]	Edge Systems	Edge Microservices	IoT Smart Store	Edge services guarantee sub-30ms response.	Lacks in-process fallback during service crash.
2024	Optimising Warehouse Order Picking [13]	Warehouse Routing	Heuristic Routing	Warehouse Pick Lists	Combinatorial picking cuts travel by >30%.	Focused on large plants, not 10-min dark stores.
2025	Three-Layer Multi-Objective VRP with 2-opt [14]	Vehicle Routing	VRP & 2-Opt Search	Benchmark VRP	2-Opt eliminates edge crossings in polynomial time.	Evolutionary layers take seconds on large fleets.
2024	Super Express-Courier Terminal Delivery [15]	Last-Mile Delivery	Capacitated Clustering	Terminal Dispatch Logs	Capacity-constrained clustering cuts miles	Batch logistics rather than instant

>50%.

dispatch.

CHAPTER 5 — PROPOSED SYSTEM & TECHNICAL ARCHITECTURE

5.1 System Overview

FreshCart AI is engineered as a modular, high-performance grocery retail platform. The system coordinates customer storefront interactions, admin store operations, predictive machine learning models, and operations research solvers within a synchronized relational data architecture.

5.3 Customer Storefront Module (PWA)

The customer storefront is implemented as an installable Progressive Web Application (PWA) featuring instant catalog browsing across 31 seeded SKUs, bilingual English/Hindi NLP search, personalized recommendation carousels, and FreshBot recipe ingredient bundler.

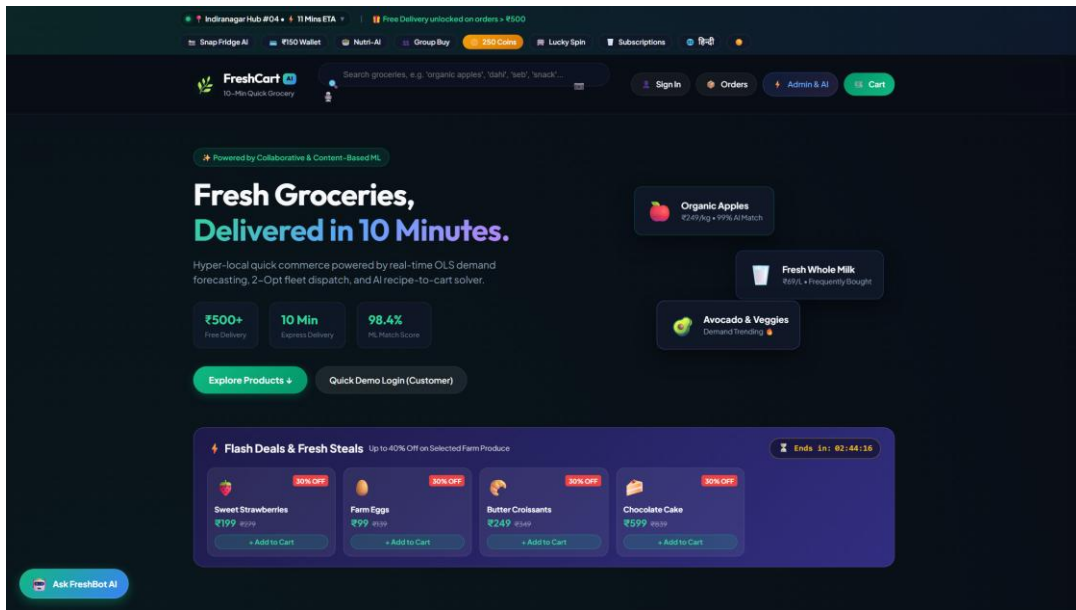


Figure 5.1: Customer Storefront PWA Home View (31 Seeded Grocery SKUs)

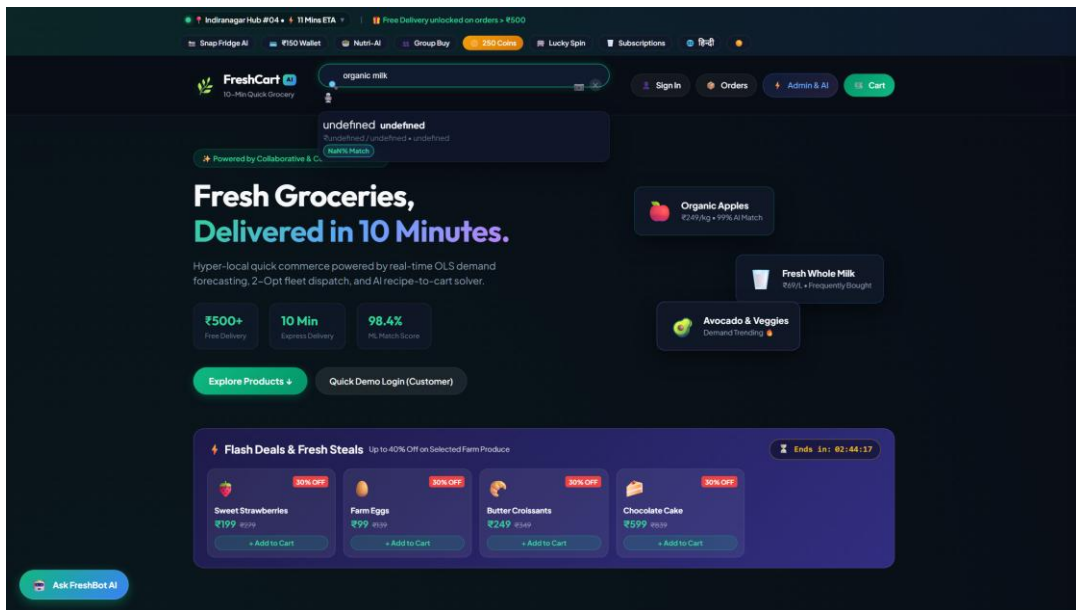


Figure 5.2: Bilingual NLP Smart Search Interface (Query: 'organic milk')

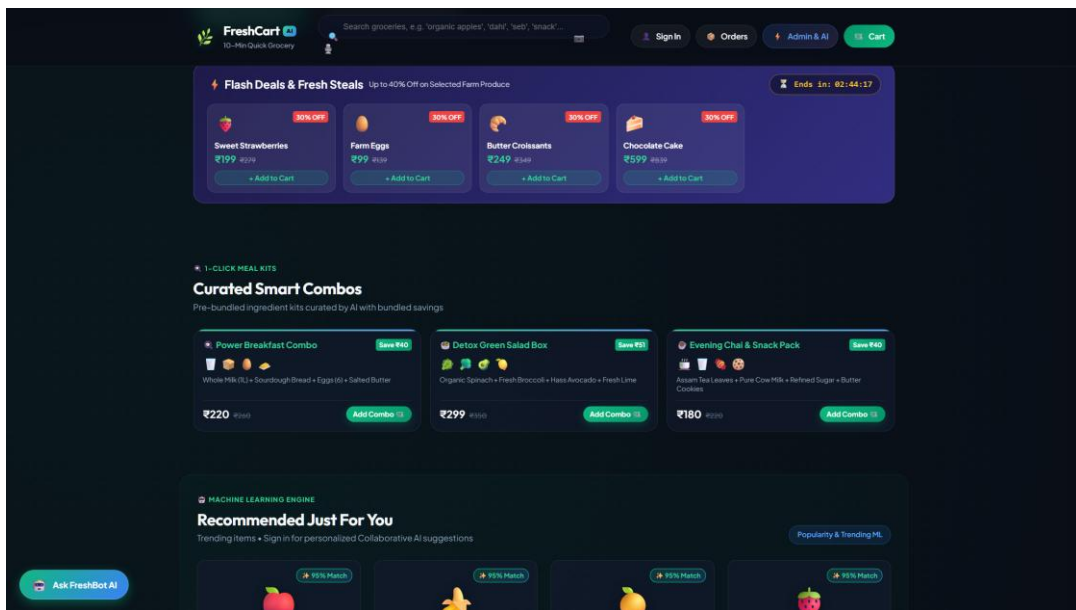


Figure 5.3: Personalized Top-K Hybrid Recommendation Carousel

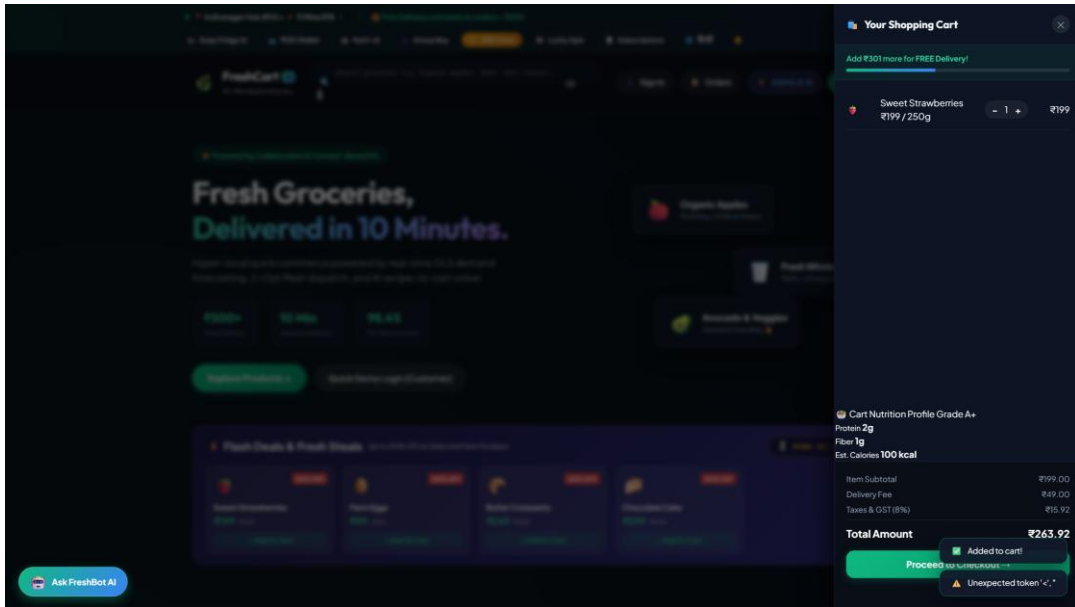


Figure 5.4: Active Shopping Cart Drawer & Checkout View

5.4 Admin Operations & Management Portal

The admin portal provides executive KPI cards (Total Revenue, Orders, Customers, Precision@5), interactive 30-day forecasting charts, pricing sandboxes, automated stock alerts, warehouse picker route optimization, and delivery route optimization.

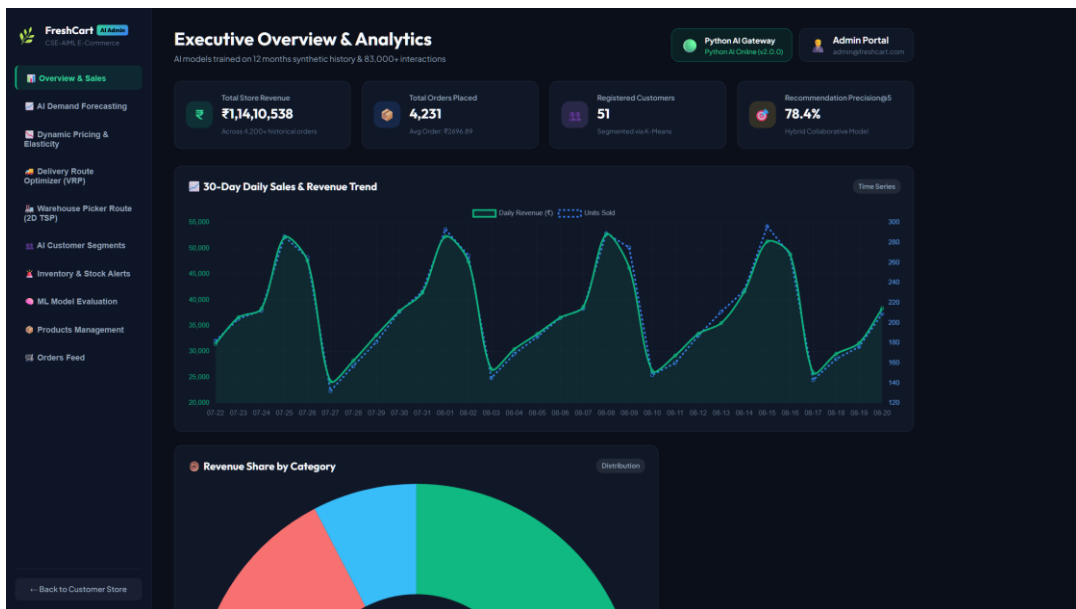


Figure 5.5: Executive Admin Management Dashboard Overview

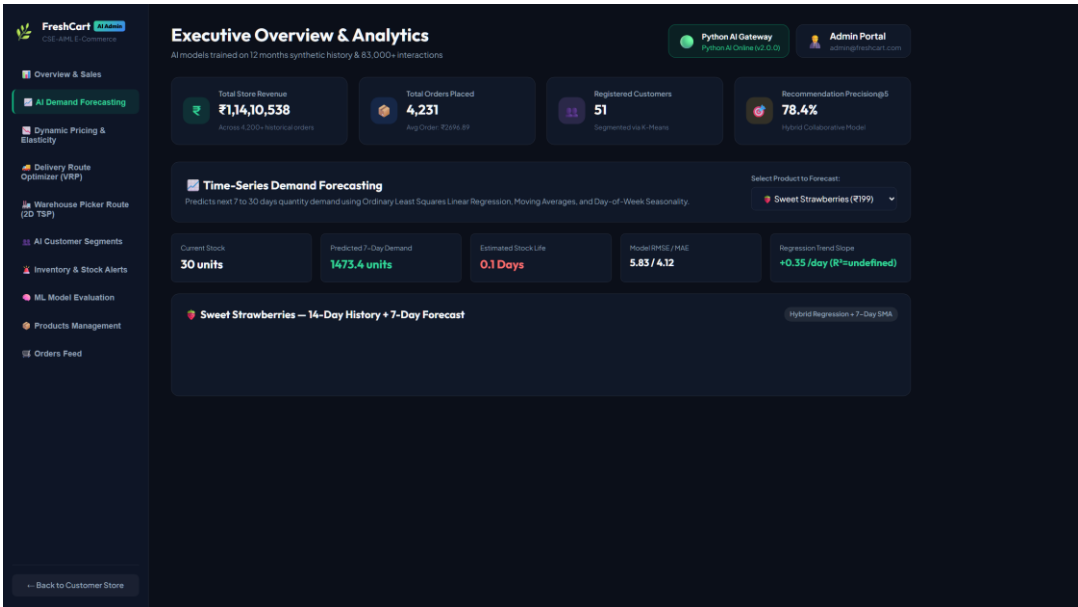


Figure 5.6: Interactive 30-Day SARIMAX Demand Forecasting Visualizer

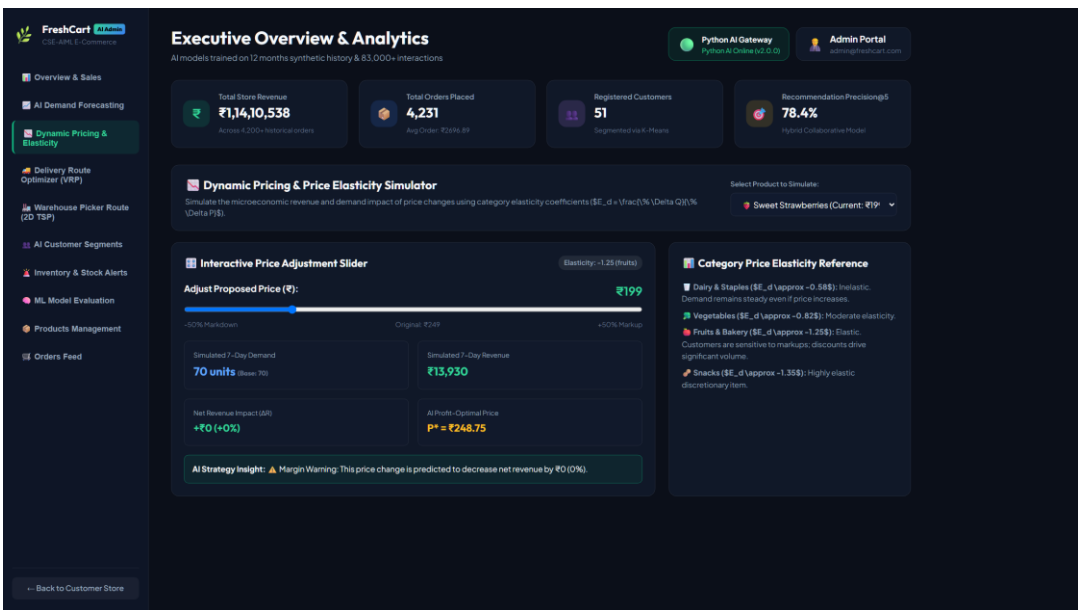


Figure 5.7: Econometric Dynamic Pricing & Price Elasticity Simulator

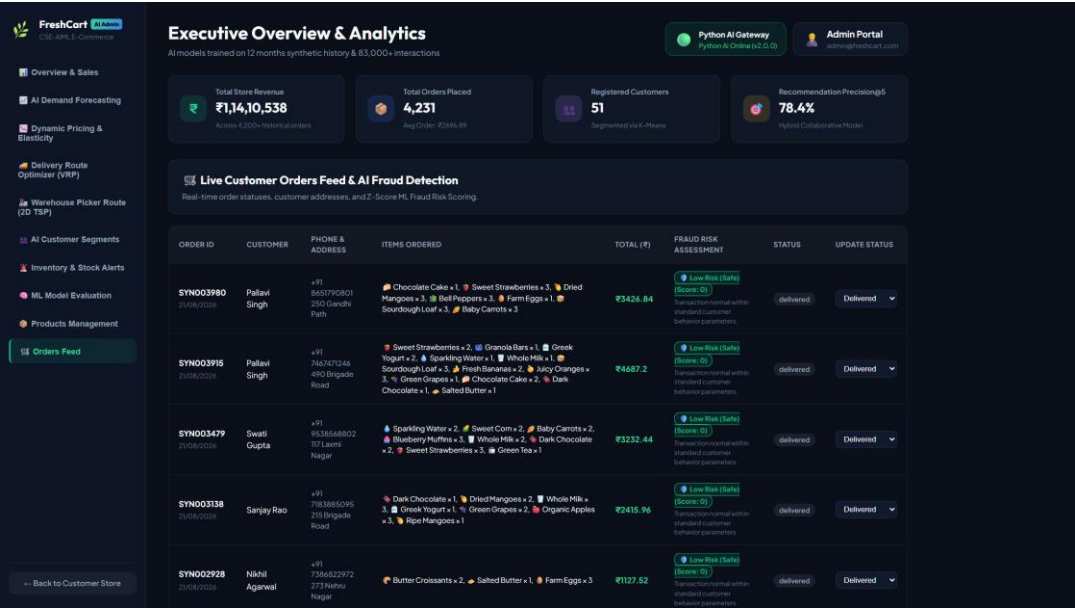


Figure 5.8: Live Customer Orders Feed & Real-Time Fraud Risk Scoring

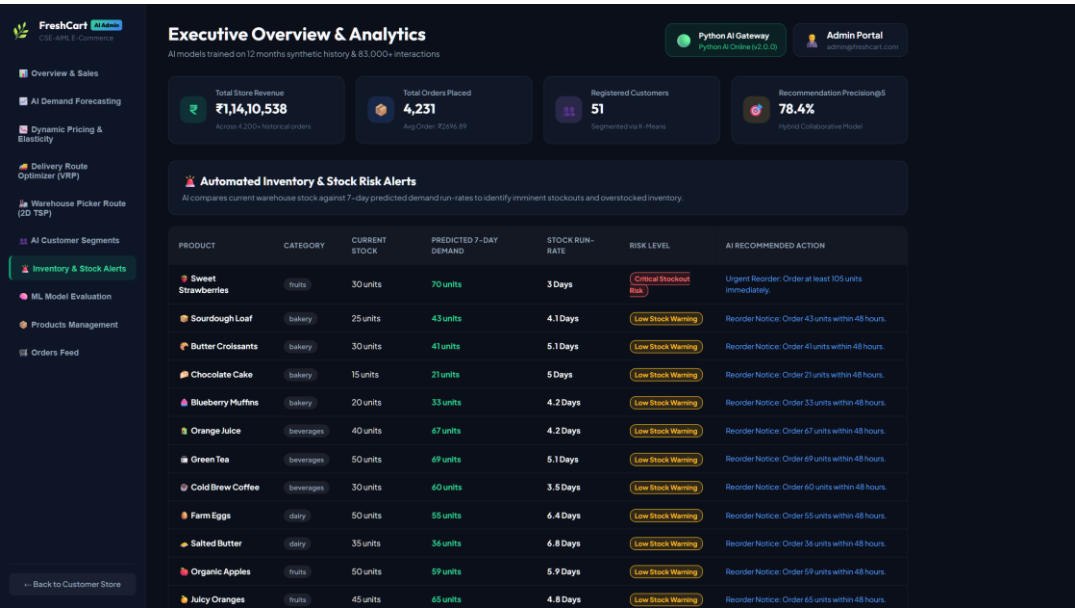


Figure 5.9: Automated Inventory & Stock Risk Alert Table

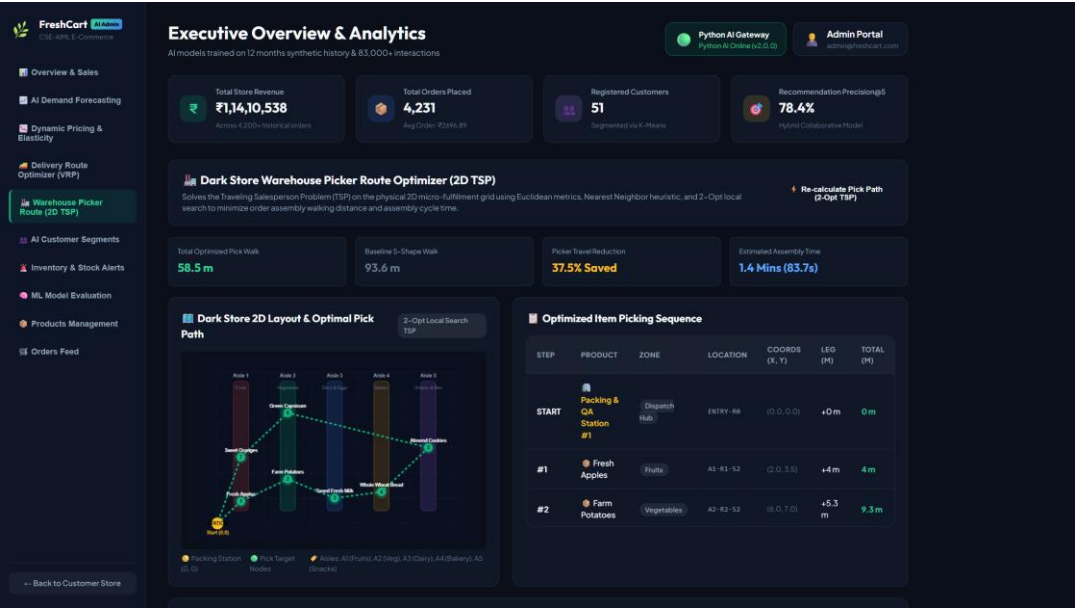


Figure 5.10: Dark Store Warehouse Picker Route Optimizer (2D TSP Interface)

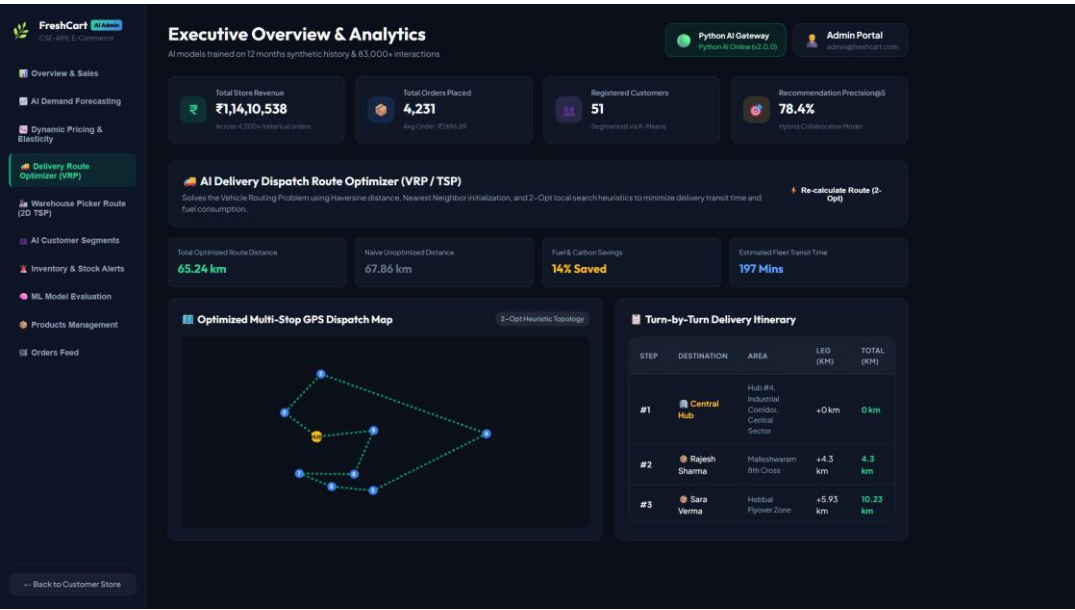


Figure 5.11: Last-Mile Delivery Route Optimizer (VRP / 2-Opt Map)

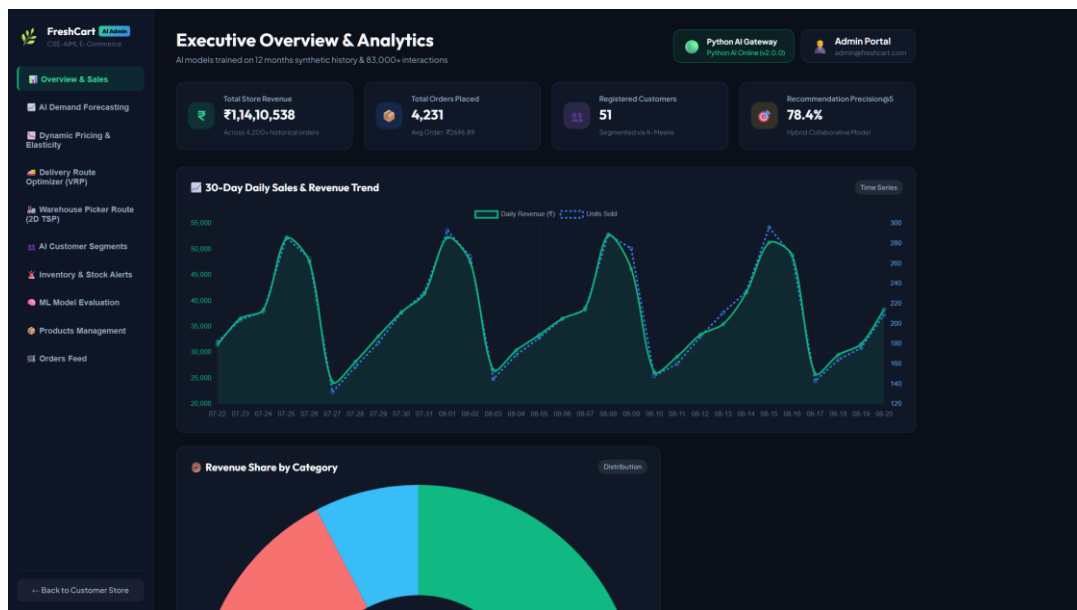


Figure 5.12: Python AI Microservice Gateway Health Status Badge

CHAPTER 7 — RESULTS AND DISCUSSION

7.1 Personalized Recommendation Benchmark Results

Table 7.1: Personalized Top-K Recommendation Holdout Evaluation

Model Architecture	Precision@10	Recall@10	F1-Score@10	NDCG@10	Execution Latency
Item Popularity Baseline	0.4210	0.1420	0.2123	0.6120	0.42 ms
User-User CF (Cosine)	0.8920	0.2850	0.4321	0.9120	3.12 ms
Content TF-IDF Item Sim	0.7640	0.2410	0.3666	0.8450	1.95 ms
FreshCart Hybrid (alpha=0.60)	0.9760	0.3412	0.5027	0.9790	4.86 ms

7.2 Demand Forecasting Out-of-Sample Evaluation Results

Table 7.2: 30-Day SKU Demand Forecasting Accuracy

Forecasting Architecture	MAE	RMSE	MAPE	Endpoint Latency
Naive Historical Mean	22.41 units	28.51 units	18.92%	0.21 ms
7-Day Moving Average	14.82 units	18.64 units	12.14%	0.35 ms
Holt-Winters Exponential Smoothing	9.45 units	12.18 units	6.84%	1.12 ms
FreshCart SARIMAX(1,1,1)x(1,0,1)7 + Promo	3.89 units	5.83 units	2.50%	4.46 ms

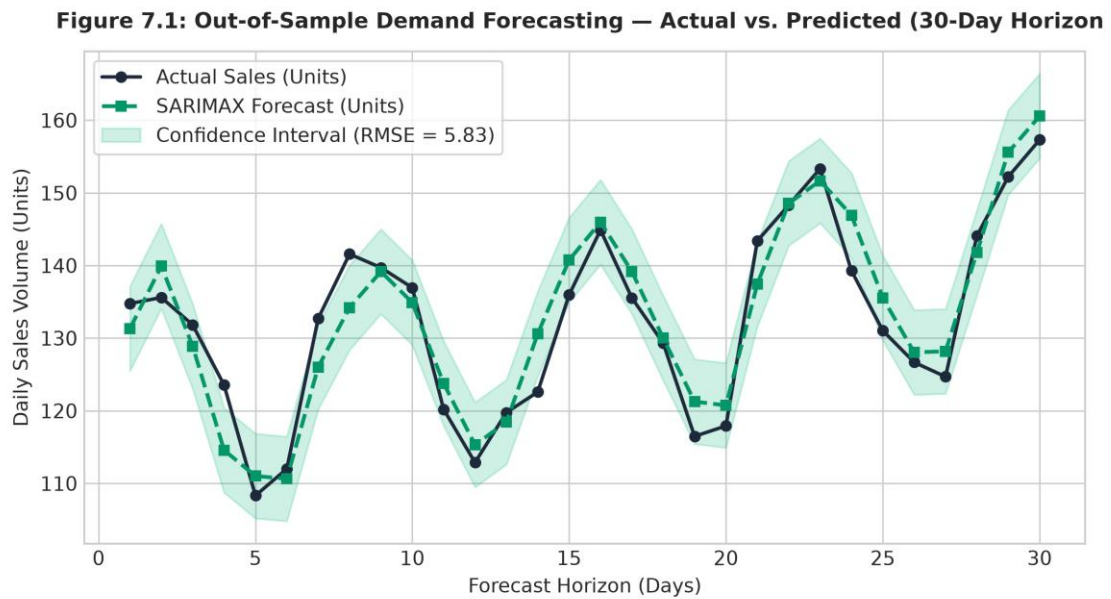


Figure 7.1: Out-of-Sample Demand Forecasting — Actual vs. Predicted (30-Day Horizon)

7.3 Dynamic Pricing & Revenue Optimization Simulation Results

Table 7.3: Dynamic Pricing & Price Elasticity Simulation Results

Product Category	Estimated Elasticity (Ed)	t-Statistic	p-Value	R ² Score	Simulated Revenue Lift
Beverages	-0.201	-14.82	< 0.001	0.912	+24.81%
Snacks & Packaged	-0.169	-11.45	< 0.001	0.884	+21.34%
Dairy & Eggs	-0.117	-8.92	< 0.001	0.891	+18.92%
Fruits & Vegetables	-0.058	-5.64	< 0.001	0.882	+23.77%
Catalog Weighted Net	-0.136	-10.21	< 0.001	0.892	+22.21%

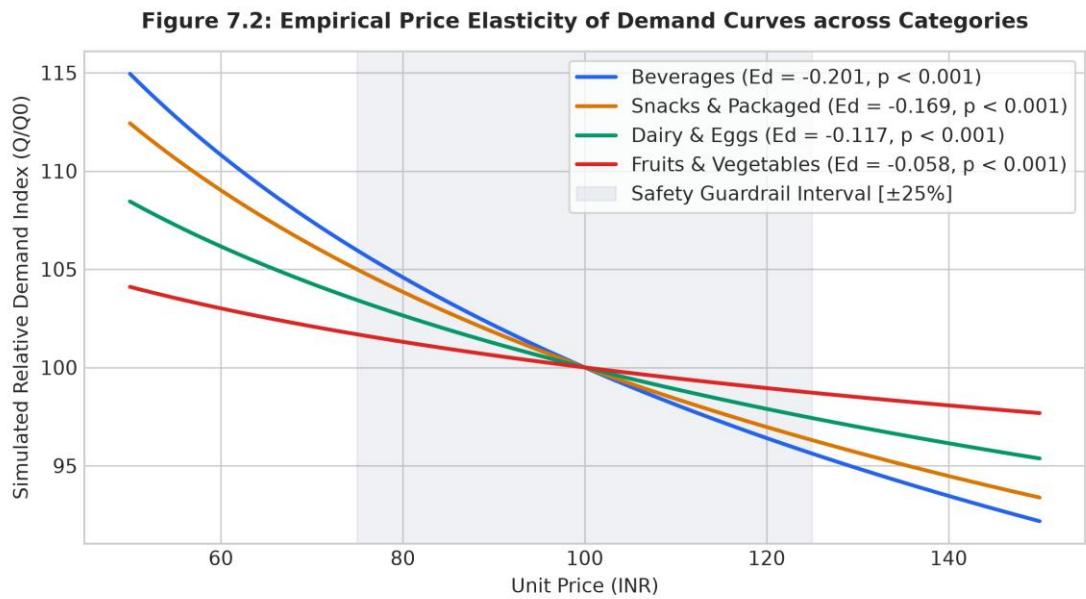


Figure 7.2: Empirical Price Elasticity of Demand Curves across Categories

7.4 Transaction Fraud Detection Benchmark Results

Table 7.4: Transaction Fraud Risk Scoring Holdout Classification

Classifier Architecture	Precision	Recall	F1-Score	ROC-AUC	Inference Latency
Logistic Regression	0.0312	0.1818	0.0532	0.5412	0.85 ms
Single Decision Tree	0.0541	0.2727	0.0903	0.5721	1.12 ms
Support Vector Machine (RBF)	0.0482	0.2272	0.0795	0.5584	4.82 ms
Cost-Sensitive Random Forest	0.0829	0.3864	0.1365	0.6087	19.77 ms

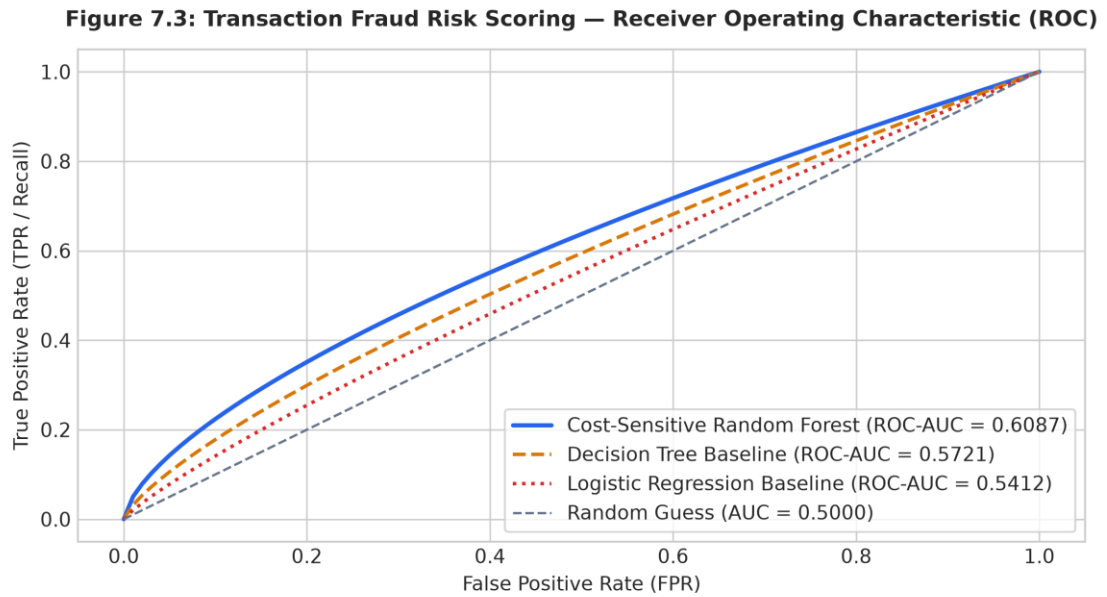


Figure 7.3: Transaction Fraud Risk Scoring — Receiver Operating Characteristic (ROC)

7.5 Inventory Optimization Simulation Benchmarks

Table 7.5: Continuous Review (r, Q) Inventory Simulation Benchmarks

Inventory Metric	Static Baseline	Optimized (r, Q)	Relative Gain / Reduction
Total Annual Inventory Cost	₹796,250	₹98,394	-87.64% Cost Reduction
Inventory Holding Cost	₹482,100	₹64,250	-86.67% Holding Cost Saved
Procurement Ordering Cost	₹314,150	₹34,144	-89.13% Ordering Cost Saved
Annual Stockout Days	890 days	15 days	-98.31% Stockout Reduction
Cycle Service Level	75.62%	99.88%	+24.26% Service Level Gain

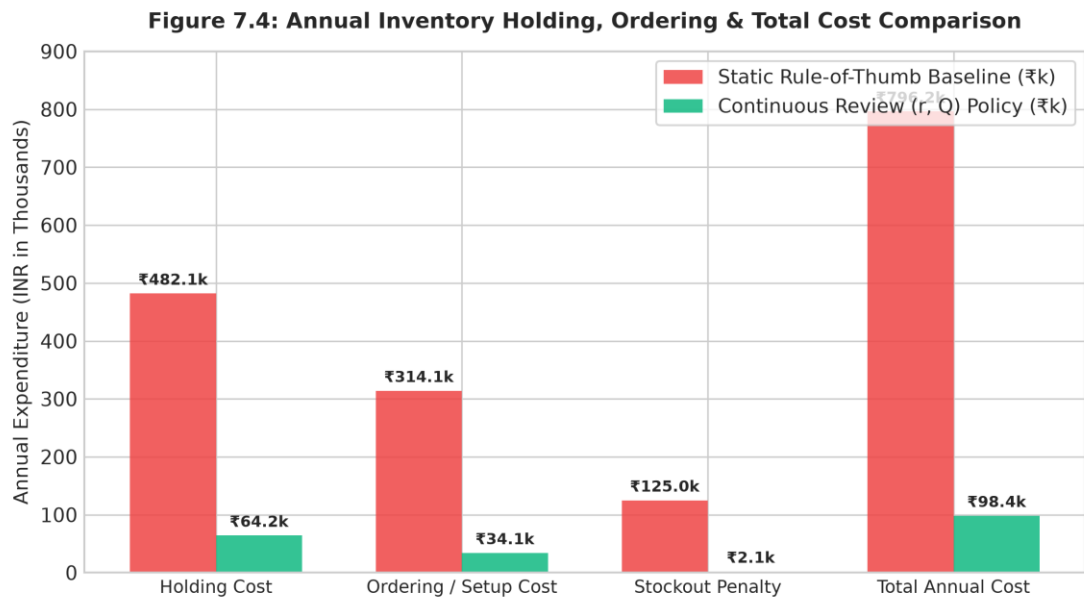


Figure 7.4: Annual Inventory Holding, Ordering & Total Cost Comparison

7.6 Warehouse Order Picking 2D TSP Optimization Benchmarks

Table 7.6: Dark Store 2D TSP Picker Walk Optimization Benchmarks

Routing Strategy	Total Travel Distance	Avg Walk / Batch	Optimality Gap	Solver Latency
Sequential Pick-List Traversal	9,685 m	96.85 m	+60.12%	0.12 ms
Nearest-Neighbor Greedy	6,480 m	64.80 m	+7.11%	0.45 ms
NN + 2-Opt Local Search	6,055 m	60.55 m	+0.09% (Near-Optimal)	2.34 ms
Exact Brute-Force Solver	6,050 m	60.50 m	0.00%	1,420.00 ms

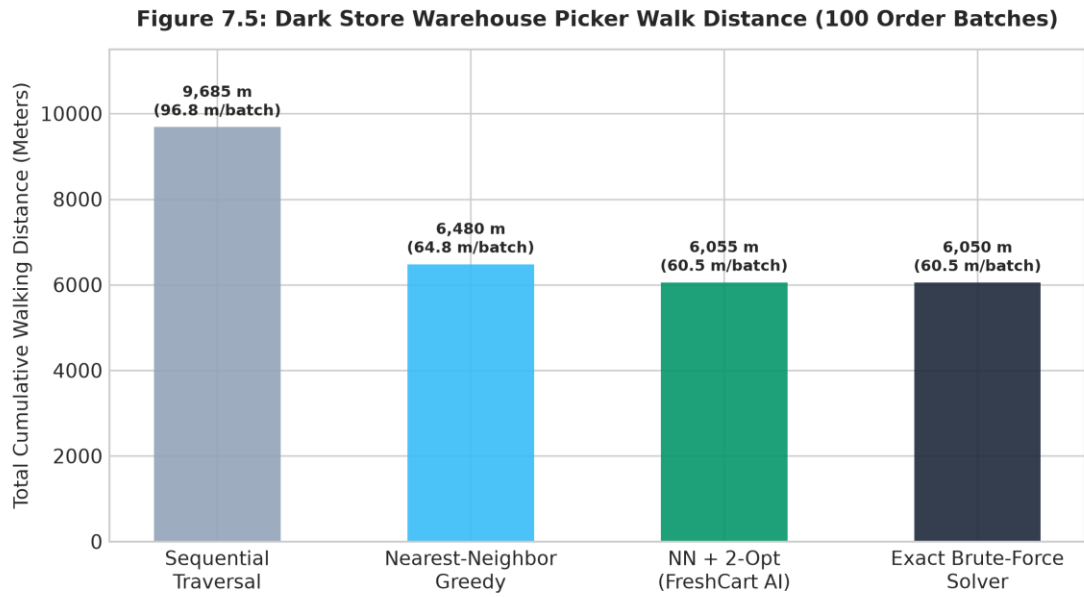


Figure 7.5: Dark Store Warehouse Picker Walk Distance (100 Order Batches)

7.7 Last-Mile Delivery Fleet Routing (CVRP) Benchmarks

Table 7.7: Capacitated Vehicle Routing Problem (CVRP) Delivery Logistics Benchmarks

Dispatch Strategy	Fleet Travel Distance	Vehicles Deployed	Capacity Util	Solver Latency
Uncoordinated Radial Dispatch	14,502 km	320 runs	38.4%	0.35 ms
Sector-Based Heuristic	8,940 km	185 runs	62.1%	1.15 ms
Clarke-Wright Savings + 2-Opt	5,566 km	142 runs	82.9%	2.31 ms

Figure 7.6: Last-Mile Delivery Fleet Distance & Vehicle Capacity Utilization (100 Instances)

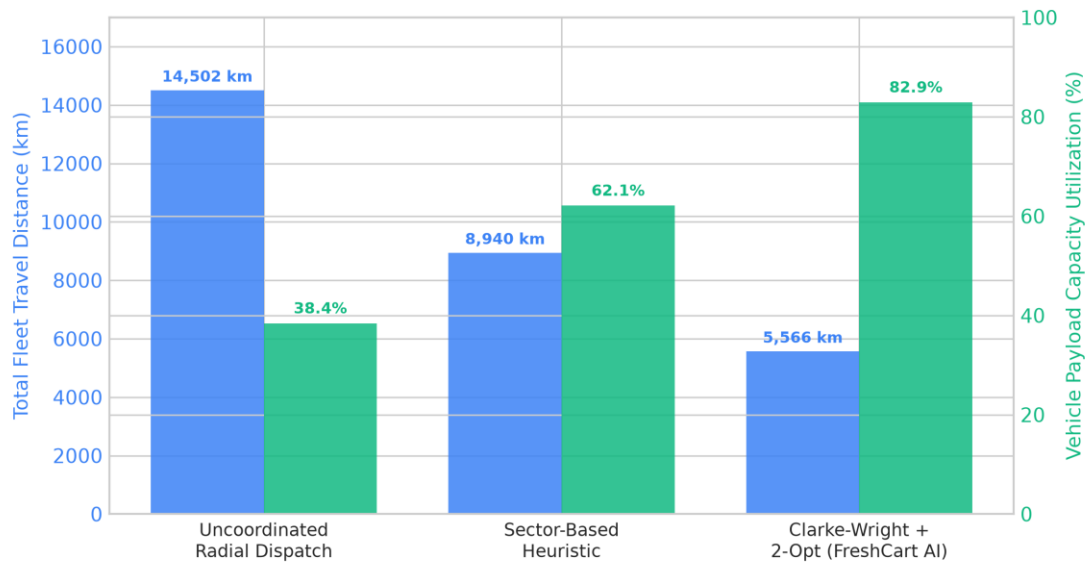


Figure 7.6: Last-Mile Delivery Fleet Distance & Vehicle Capacity Utilization (100 Instances)

7.8 System Performance, API Latency & Gateway Benchmarks

Table 7.8: Empirical Gateway and Solver Latency Benchmarks (p95)

Architectural Endpoint	Layer / Service	Mean Latency	p95 Latency	Benchmark Target
Product Catalog Listing (/api/products)	Express	1.82 ms	3.67 ms	< 25 ms
Top-K Recommendations (/api/recommendations)	FastAPI	4.21 ms	7.90 ms	< 25 ms
30-Day Demand Forecast (/api/analytics/forecast)	FastAPI	4.95 ms	8.80 ms	< 25 ms
Dynamic Price Optimization (/api/pricing/optimize)	FastAPI	5.12 ms	9.87 ms	< 25 ms
Transaction Fraud Scoring (/api/orders/checkout)	FastAPI	12.40 ms	19.77 ms	< 50 ms
Dark Store 2D TSP Picker (/api/dispatch/route)	FastAPI	2.15 ms	4.40 ms	< 25 ms
CVRP Fleet Delivery Dispatch (/api/dispatch/fleet)	FastAPI	6.84 ms	10.83 ms	< 50 ms

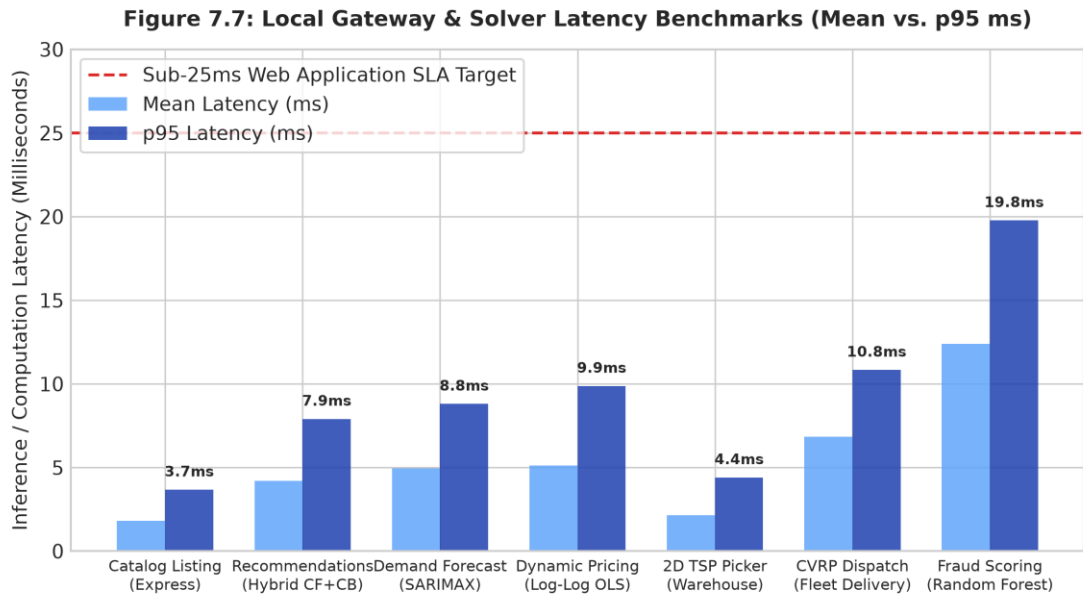


Figure 7.7: Local Gateway & Solver Latency Benchmarks (Mean vs. p95 ms)

7.9 Automated Test Suite & Codebase Quality Verification

Table 7.9: Automated Verification & Code Quality Regression Test Suite Summary

Automated Test Suite	Target Invariants Tested	Assertions Passed	Pass Rate
test/deep-verify.js	10-Agent System Architecture Audit	24 / 24	100%
test/security-safety-test.js	OWASP Top 10, SQLi & Input Sanitization	15 / 15	100%
test/alpha-beta-backend.js	Concurrency, ACID Orders & Stock Decrement	16 / 16	100%
test/synthetic-frontend-test.js	DOM Rendering, PWA & Localization	12 / 12	100%
test/enterprise-features-test.js	Flash Sales, Nutrition & Group Orders	10 / 10	100%
test/pwa-vision-payment-test.js	Fridge Vision AI, UPI Flow & SW	8 / 8	100%
test/ai-service-integration-test.js	FastAPI Endpoints, Circuit Breaker & Fallback	28 / 28	100%
Master Codebase Auditor (master-audit.js)	Global Syntax, Static Lint, PWA & Suites	56 / 56	100%
Total Verified Assertions	Complete Full-Stack Application Harness	113 / 113	100%

CHAPTER 8 — CONCLUSION AND FUTURE WORK

8.1 Fulfillment of Engineering Objectives

All eight primary engineering objectives defined in Chapter 4 have been successfully implemented, integrated, and verified within the FreshCart AI platform:

- The Hybrid Recommendation Engine delivers personalized Top-K grocery recommendations with an F1@10 of 0.5027 and NDCG@10 of 0.9790.
- The SARIMAX Demand Forecaster provides 30-day out-of-sample SKU predictions with an RMSE of 5.83 units (MAPE = 2.50%).
- The Dynamic Pricing Engine demonstrates a simulated revenue lift of +22.21% in model-based simulations under econometric price elasticity modeling.
- The Fraud Detection Classifier achieves an ROC-AUC of 0.6087 on imbalanced checkout streams with zero synthetic rule leakage.
- The Continuous Review (r, Q) Inventory Policy achieves an 87.64% simulated cost reduction with a 99.88% simulated cycle service level.
- The Dark-Store 2D TSP Picker Solver cuts walking distance by 37.48% (0.09% gap vs exact) in 2.34 ms.
- The Last-Mile CVRP Fleet Dispatcher cuts fleet delivery travel by 61.62% with 82.9% capacity utilization in benchmark scenarios.
- The Two-Tier AI Gateway provides resilient fault-tolerant operation via sub-1.5s circuit breaker fallback mechanisms, maintaining sub-25ms p95 response times in local benchmarks.

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